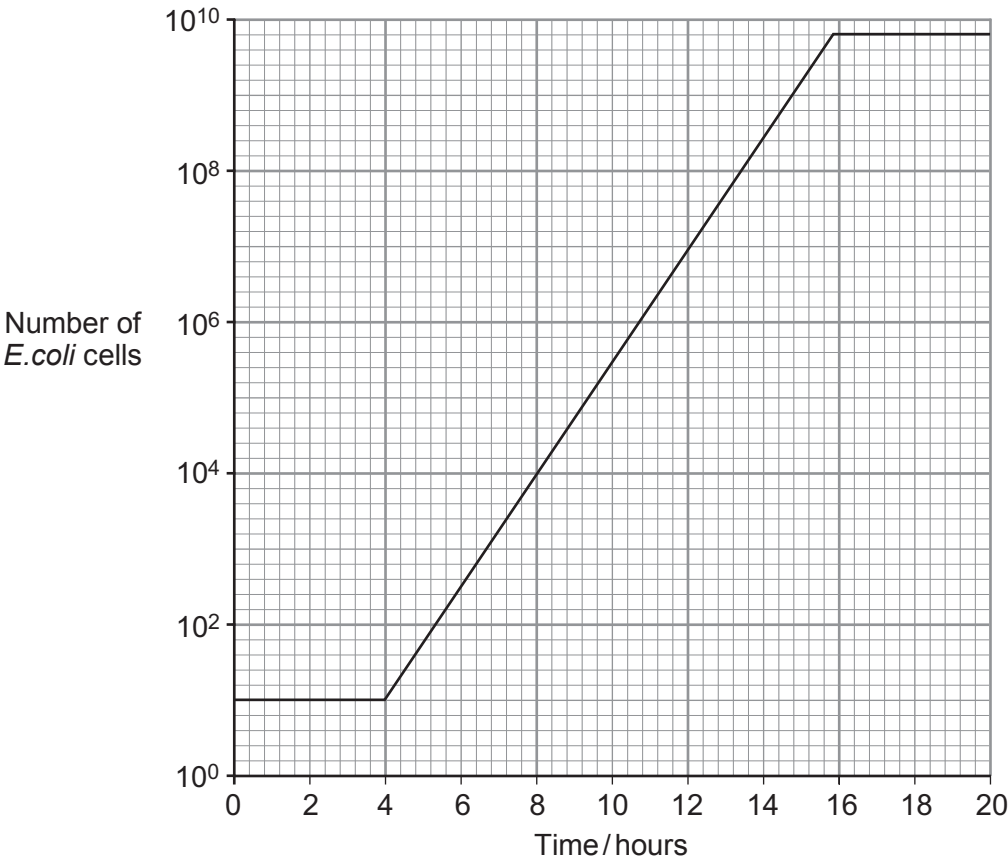


4. Bacterial populations grow exponentially due to binary fission. The generation time is the time interval required for the cells to divide. A scientist counted the number of living *Escherichia coli* bacterial cells present over a 20 hour period. She plotted the results on a graph as shown below.



- (a) Use the equation below to calculate the generation time of the bacterial population between 4 and 8 hours of growth. Give your answer to the nearest minute. [3]

$$G = \frac{t}{3.3 \left(\frac{\log b}{\log B} \right)}$$

Where:

G = generation time (minutes)

t = time interval (minutes)

B = number of bacteria at the beginning of the time interval

b = number of bacteria at the end of the time interval

time = minutes



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The scientist carried out the experiment again but counted the living bacteria present each hour for 60 hours.

- (b) (i) Describe and explain the expected shape of the population growth curve between 20 and 60 hours. [2]

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- (ii) Explain how the growth curve would differ if a total count had been used to measure the population density. Suggest a disadvantage of this technique. [2]

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